A transforms

bins=32

Part A — Paràmetres de normalització (transforms)

Paràmetre	Valor
z_mean	-18.052
z_std	10.603
R_max	20
log_edep_mean	-4.4352
log_edep_std	2.9554
edep_offset_MeV	1e-12
log_t_mean	3.6413
log_t_std	1.8725
t_offset_ns	0.001
log_E_in_mean	-2.2675
log_E_in_std	3.1847
z_mean_poly_coeffs	[0.1187, 1.4921, 4.8059, -20.731]
condZ	✓

bins=64

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z_mean	-18.052
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log_edep_mean	-4.4352
log_edep_std	2.9554
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log_E_in_mean	-2.2675
log_E_in_std	3.1847
z_mean_poly_coeffs	[0.1187, 1.4921, 4.8059, -20.731]
condZ	✓

bins=16+200k

Part A — Paràmetres de normalització (transforms)

Paràmetre	Valor
z_mean	-18.052
z_std	10.603
R_max	20
log_edep_mean	-4.4352
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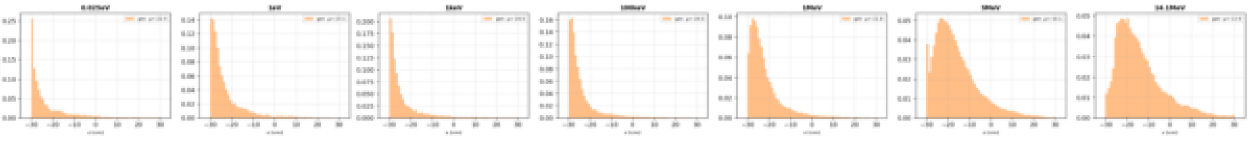
t_offset_ns	0.001
log_E_in_mean	-2.2675
log_E_in_std	3.1847
z_mean_poly_coeffs	[0.1187, 1.4921, 4.8059, -20.731]
condZ	✓

C z phys

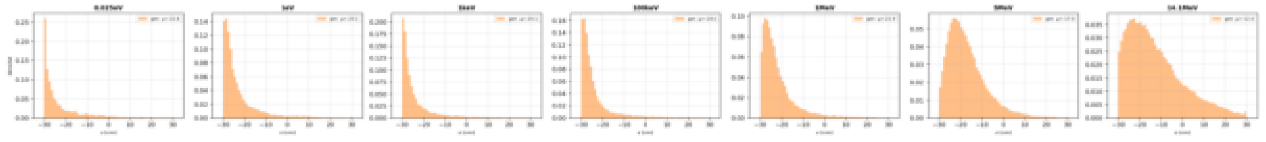
bins=32

Part C — Distribució z física [cm] (truth vs generated)
Blau = truth (Geant4) Taronja = generated (EPIC-FM)



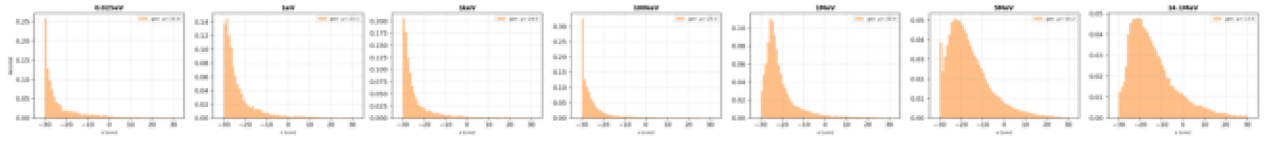
bins=64

Part C — Distribució z física [cm] (truth vs generated)
Blau = truth (Geant4) Taronja = generated (EPIC-FM)



bins=16+200k

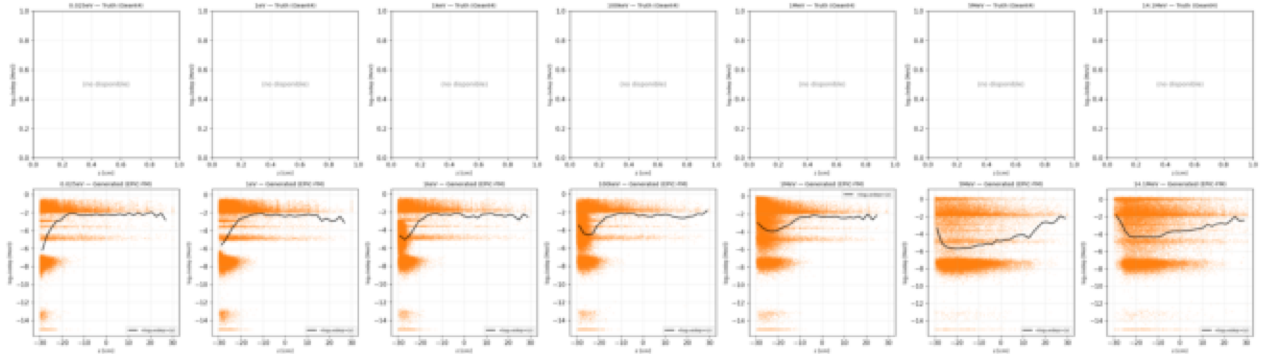
Part C — Distribució z física [cm] (truth vs generated)
Blau = truth (Geant4) Taronja = generated (EPIC-FM)



D edep vs z scatter

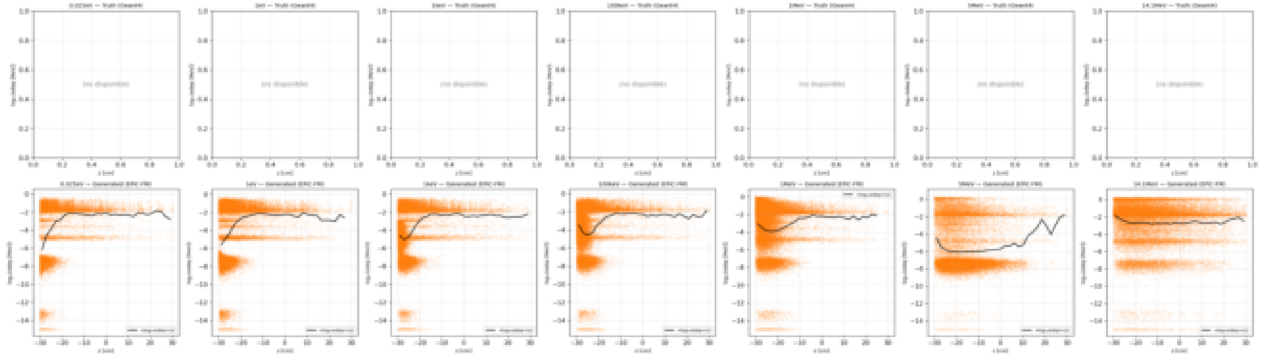
bins=32

Part D — Scatter edep vs z per hit
Fila dalt = truth   Fila baix = generated



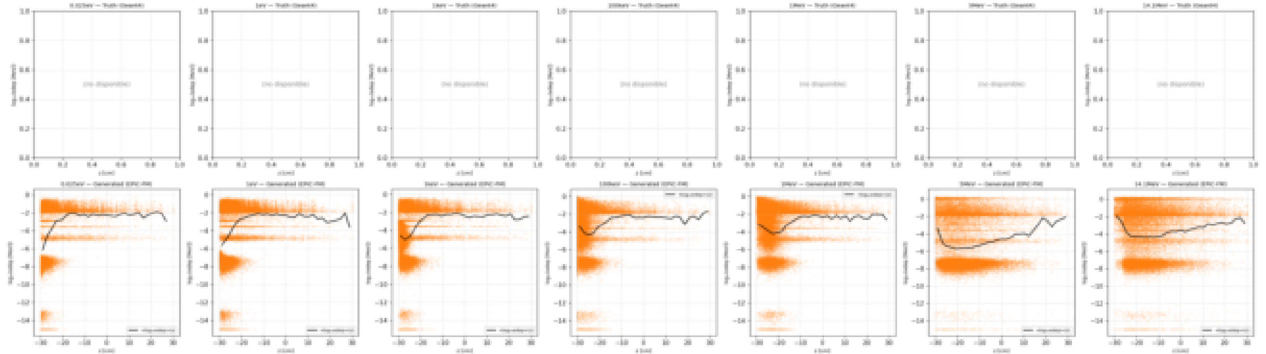
bins=64

Part D — Scatter edep vs z per hit
Fila dalt = truth   Fila baix = generated



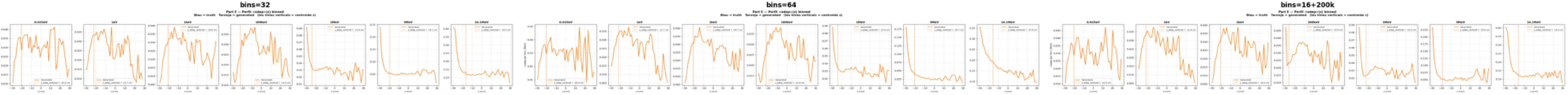
bins=16+200k

Part D — Scatter edep vs z per hit
Fila dalt = truth   Fila baix = generated

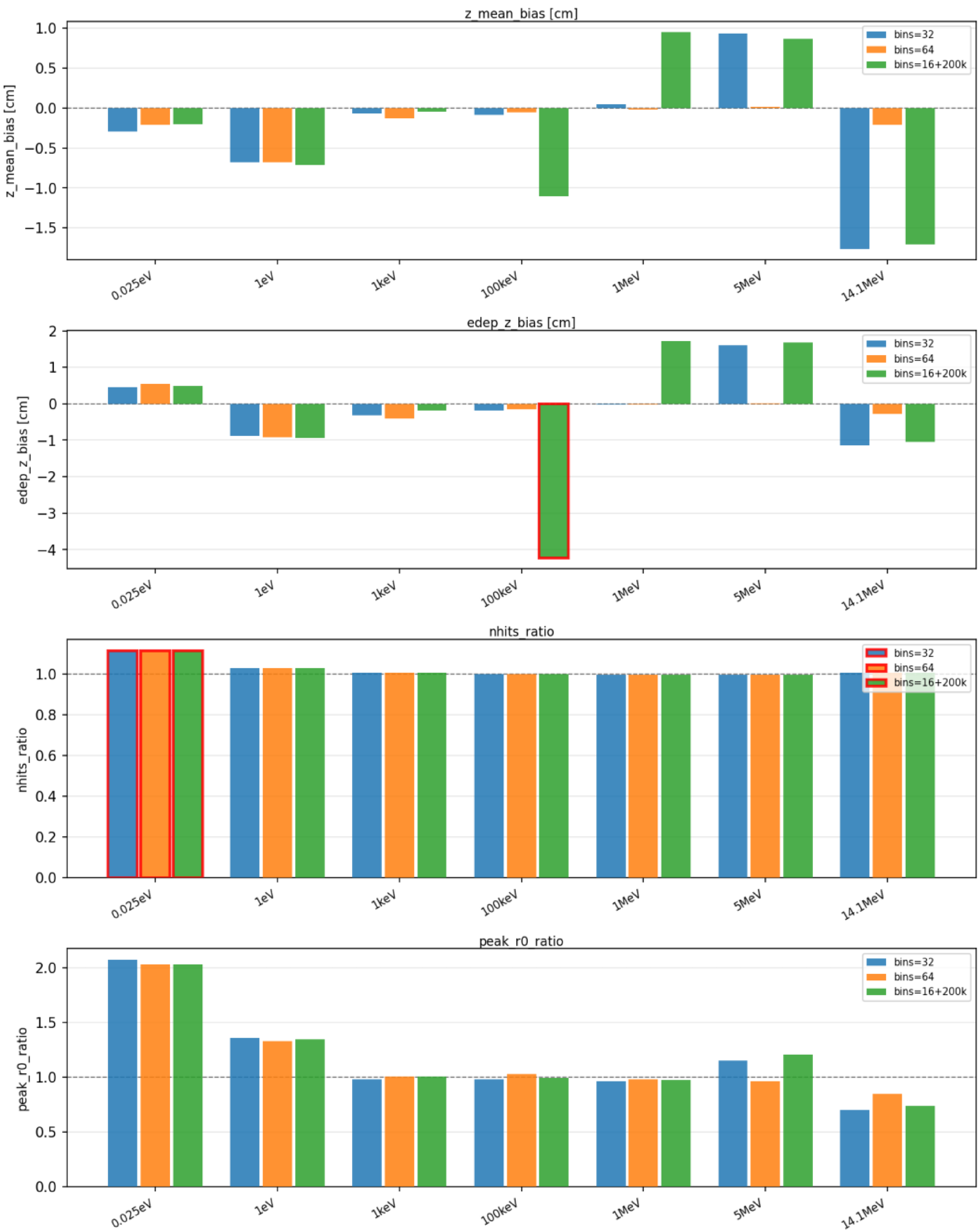


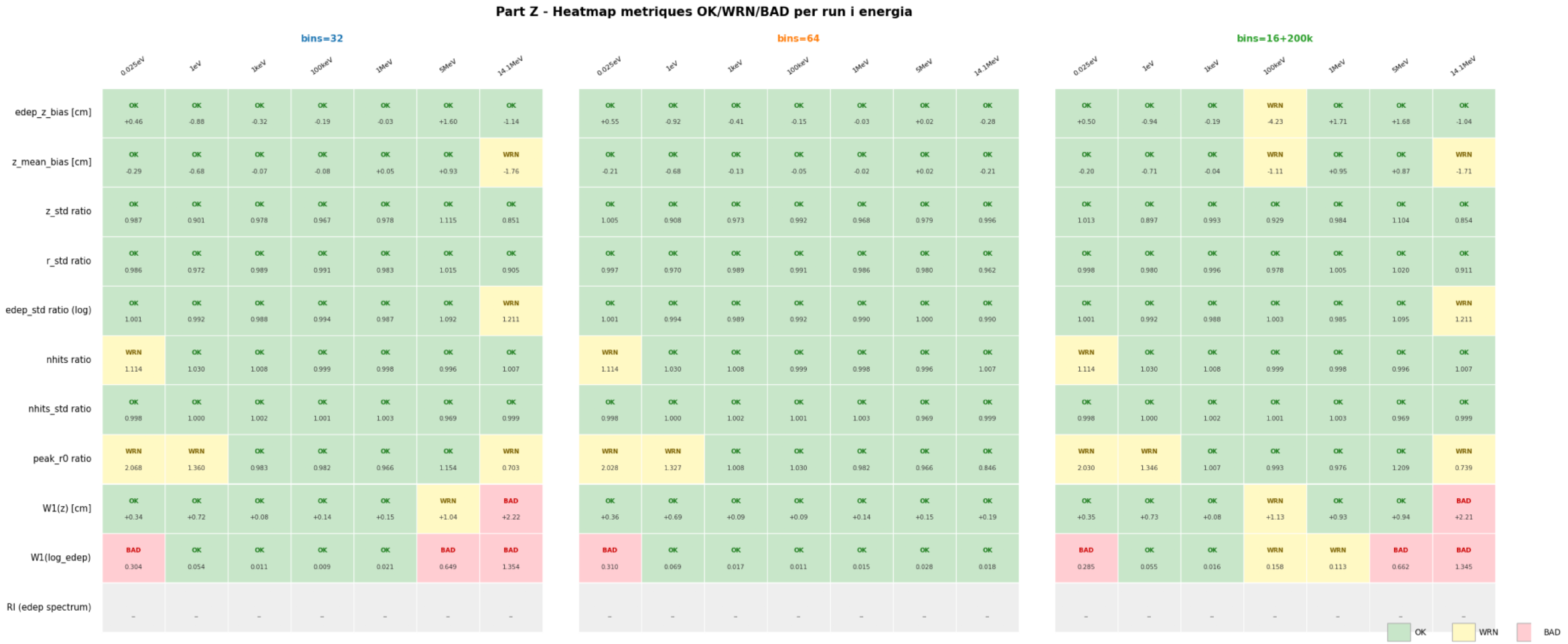
```
Bandes horitzontals esperades (neutrons en CH2):
  log10(edep) ~ +0.65  |  12C(n,n'g) inelastic, g = 4.44 MeV
  log10(edep) ~ +0.35  |  H(n,g)D   captura termica H, g = 2.22 MeV
  log10(edep) ~ +0.10  |  12C(n,g)  captura C, g = 1.26 MeV
  continu               |  n(H,H)n   recul de proto (elastic), E in [0, En]
  continu               |  Compton / fotoelectric de g secundaris
  log10(edep) ~ -14    |  Floor de tracking Geant4 (electrons baixa E)
```

E edep z profile



Part M - Resum metriques per run i energia




Part Z - Heatmap metriques OK/WRN/BAD per run i energia
bins=32
bins=64
bins=16+200k
0.025eV 1eV 1keV 100keV 1MeV 5MeV 14.1MeV
edep_z_bias [cm]
z_mean_bias [cm]
z_std ratio
r_std ratio
edep_std ratio (log)
nhits ratio
nhits_std ratio
peak_r0 ratio
W1(z) [cm]
W1(log_edep)
RI (edep spectrum)
bins=32:
edep_z_bias [cm]: OK +0.46 | OK -0.88 | OK -0.32 | OK -0.19 | OK -0.03 | OK +1.60 | OK -1.14
z_mean_bias [cm]: OK -0.29 | OK -0.68 | OK -0.07 | OK -0.08 | OK +0.05 | OK +0.93 | WRN -1.76
z_std ratio: OK 0.987 | OK 0.901 | OK 0.978 | OK 0.967 | OK 0.978 | OK 1.115 | OK 0.851
r_std ratio: OK 0.986 | OK 0.972 | OK 0.989 | OK 0.991 | OK 0.983 | OK 1.015 | OK 0.905
edep_std ratio (log): OK 1.001 | OK 0.992 | OK 0.988 | OK 0.994 | OK 0.987 | OK 1.092 | WRN 1.211
nhits ratio: WRN 1.114 | OK 1.030 | OK 1.008 | OK 0.999 | OK 0.998 | OK 0.996 | OK 1.007
nhits_std ratio: OK 0.998 | OK 1.000 | OK 1.002 | OK 1.001 | OK 1.003 | OK 0.969 | OK 0.999
peak_r0 ratio: WRN 2.068 | WRN 1.360 | OK 0.983 | OK 0.982 | OK 0.966 | OK 1.154 | WRN 0.703
W1(z) [cm]: OK +0.34 | OK +0.72 | OK +0.08 | OK +0.14 | OK +0.15 | WRN +1.04 | BAD +2.22
W1(log_edep): BAD 0.304 | OK 0.054 | OK 0.011 | OK 0.009 | OK 0.021 | BAD 0.649 | BAD 1.354
RI (edep spectrum): – | – | – | – | – | – | –
bins=64:
edep_z_bias [cm]: OK +0.55 | OK -0.92 | OK -0.41 | OK -0.15 | OK -0.03 | OK +0.02 | OK -0.28
z_mean_bias [cm]: OK -0.21 | OK -0.68 | OK -0.13 | OK -0.05 | OK -0.02 | OK +0.02 | OK -0.21
z_std ratio: OK 1.005 | OK 0.908 | OK 0.973 | OK 0.992 | OK 0.968 | OK 0.979 | OK 0.996
r_std ratio: OK 0.997 | OK 0.970 | OK 0.989 | OK 0.991 | OK 0.986 | OK 0.980 | OK 0.962
edep_std ratio (log): OK 1.001 | OK 0.994 | OK 0.989 | OK 0.992 | OK 0.990 | OK 1.000 | OK 0.990
nhits ratio: WRN 1.114 | OK 1.030 | OK 1.008 | OK 0.999 | OK 0.998 | OK 0.996 | OK 1.007
nhits_std ratio: OK 0.998 | OK 1.000 | OK 1.002 | OK 1.001 | OK 1.003 | OK 0.969 | OK 0.999
peak_r0 ratio: WRN 2.028 | WRN 1.327 | OK 1.008 | OK 1.030 | OK 0.982 | OK 0.966 | OK 0.846
W1(z) [cm]: OK +0.36 | OK +0.69 | OK +0.09 | OK +0.09 | OK +0.14 | OK +0.15 | OK +0.19
W1(log_edep): BAD 0.310 | OK 0.069 | OK 0.017 | OK 0.011 | OK 0.015 | OK 0.028 | OK 0.018
RI (edep spectrum): – | – | – | – | – | – | –
bins=16+200k:
edep_z_bias [cm]: OK +0.50 | OK -0.94 | OK -0.19 | WRN -4.23 | OK +1.71 | OK +1.68 | OK -1.04
z_mean_bias [cm]: OK -0.20 | OK -0.71 | OK -0.04 | WRN -1.11 | OK +0.95 | OK +0.87 | WRN -1.71
z_std ratio: OK 1.013 | OK 0.897 | OK 0.993 | OK 0.929 | OK 0.984 | OK 1.104 | OK 0.854
r_std ratio: OK 0.998 | OK 0.980 | OK 0.996 | OK 0.978 | OK 1.005 | OK 1.020 | OK 0.911
edep_std ratio (log): OK 1.001 | OK 0.992 | OK 0.988 | OK 1.003 | OK 0.985 | OK 1.095 | WRN 1.211
nhits ratio: WRN 1.114 | OK 1.030 | OK 1.008 | OK 0.999 | OK 0.998 | OK 0.996 | OK 1.007
nhits_std ratio: OK 0.998 | OK 1.000 | OK 1.002 | OK 1.001 | OK 1.003 | OK 0.969 | OK 0.999
peak_r0 ratio: WRN 2.030 | WRN 1.346 | OK 1.007 | OK 0.993 | OK 0.976 | OK 1.209 | WRN 0.739
W1(z) [cm]: OK +0.35 | OK +0.73 | OK +0.08 | WRN +1.13 | OK +0.93 | OK +0.94 | BAD +2.21
W1(log_edep): BAD 0.285 | OK 0.055 | OK 0.016 | WRN 0.158 | WRN 0.113 | BAD 0.662 | BAD 1.345
RI (edep spectrum): – | – | – | – | – | – | –
OK
WRN
BAD